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Slot-B

# CO3 - ASSESSMENT - Tool-3

1.) Task 1 - Context - Free Grammar Design and Derivation

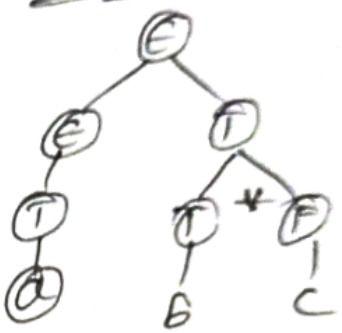
1.) Definition

A context free Grammar (CFG) is used to describe the syntax of a language. It consists of terminal non terminals Production symbol.

2.) Given expression

a+b*c

Parse tree




3.) Context free Grammar

* E → E+T/T
* T → T* F/F
* F → (E)/a/b/c

4.) Grammar components

* Terminals: +, *, (), a, b, c
* NON terminals: E, T, F
* Start symbol: E
* Production rules: Given above

5.) Leftmost Derivation

E => E+T
=> T+T
F+T
a+T
a+T*E
a+F*F
a+b*c

6.) Rightmost Derivation.

E -> E+T
E+T*E
E+T*c
E+F*c
E+b*c
T+b*c
F+b*c
a+b*c

2.) Task 2 :- Simplification of context-free Grammar

✓ S → AB/a
✓ A → aA/ε
✓ B → bB/C
✓ C → C

1.) Nullable variable

A is nullable because A → ε

2.) elimination ε-Productions

since A is nullable remove A from S → AB

✓ S → AB/B/a
✓ A → aA/a
✓ B → bB/C
✓ C → C

3.) eliminate unit Productions.

unit Prodution are S → B and B → C

Replace them using the Productions of B and C

✓ S → AB/bB/c/a
✓ A → aA/a
✓ B → bB/C

4.) Final simplification CFG

S → AB/bB/c/a
A → aA/a
B → bB/C

Conclusion:- The simplified grammar has no ε-Productions

Task 3:- Conversion into chomskey Normal form

- S → aAB/b
- A → aA/a
- B → bc/b

1.) CNF Rules

In chomsky Normal Form Production must beign the following

- A → BC
- A → a

2) Introduce New variables

Replace terminal symbol in longer Productions

- X → a
- Y → b

3.) convert Production :-

orginal Production

S → aAB/b

Replace a with x:

S → X AB/b

4.) Verification:-

All Productions are in the form A → BC or A → a ∴ the grammar satisfies Chomsky Normal form

4.) Task 4:- Conversion into Greibach Normal

- S → ABa
- A → aAa
- B → bB/b

1.) GNF Rule

In Greibach Normal Form, every Production must begin with terminal symbol by or more variables

2.) Identify Incorrect Production

S → AB does not satisfy GNF because it begins with a variable

3.) Apply Transformation

Substitute A Productions into S → AB

Since A → aAa

we get:

S → aAB/aBa

4.) Derivation of string aab

S → aAB
→ aaB
→ aab

Conclusion:- the given grammar is converted into Greibach Normal Form and generates the required string aab.

Task 5:- Context Free language Properties and Pumping lemma.

$$L = \{a^n b^n c^n / n \geq 1\}$$

1.) Context - Free language.

The given language is not context free because it requires equal number of a, b and c.

2.) Pumping lema :-

For every context free language, a sufficiently long string can be divided into five parts

3.) select a suitable string

Let p be the pumping length

$$S = a^p b^p c^p$$

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the string belongs to the given language

4.) Apply Pumping lemma

Let string belongs $|vxy| \leq p$ the substring vxy can contain symbols from at most two adjacent blocks.

Conclusion:-

L is not a context - Free language